

EEE F313: ADVD Lab 1 Report

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1 Introduction

The first lab involved learning how to access and use virtuoso, as well as create and use a basic circuit. We were required to then plot I_D against V_{DS} & V_{GS} as well as V_{GS} against V_{DS} for an NMOS and PMOS transistor in a basic circuit.

We were also required to identify V_{th} from the graphs.

2 NMOS Characteristics

The NMOS circuit is shown in figure 1.

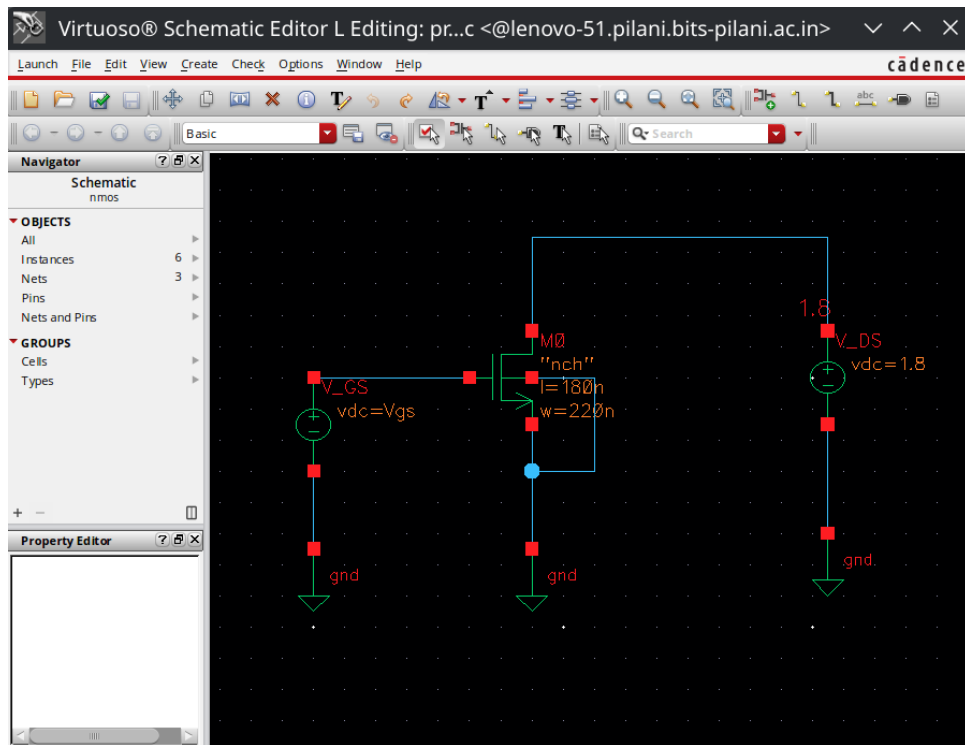


Figure 1: NMOS Circuit

The values for V_{DS} were varied between 0V and 1.8V, with a step size of 0.01V. A DC analysis was plotted, with the NMOS parameters being $l = 180nm$, $w = 220nm$, and $V_{GS} = 1.8V$.

2.1 NMOS Characteristics

Upon performing the plot, the following graph was obtained: From 2, we can see that

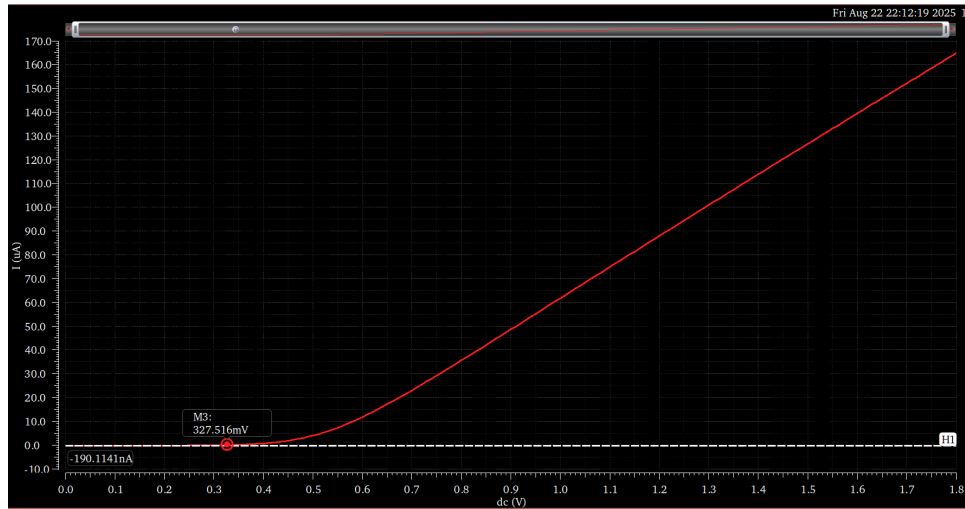


Figure 2: I_D vs V_{GS} for NMOS

the threshold voltage V_{th} is approximately $0.4V$, as this is where the current I_D starts to increase significantly.

2.2 I_D vs V_{DS} for NMOS

The following graph was obtained for I_D vs V_{DS} :

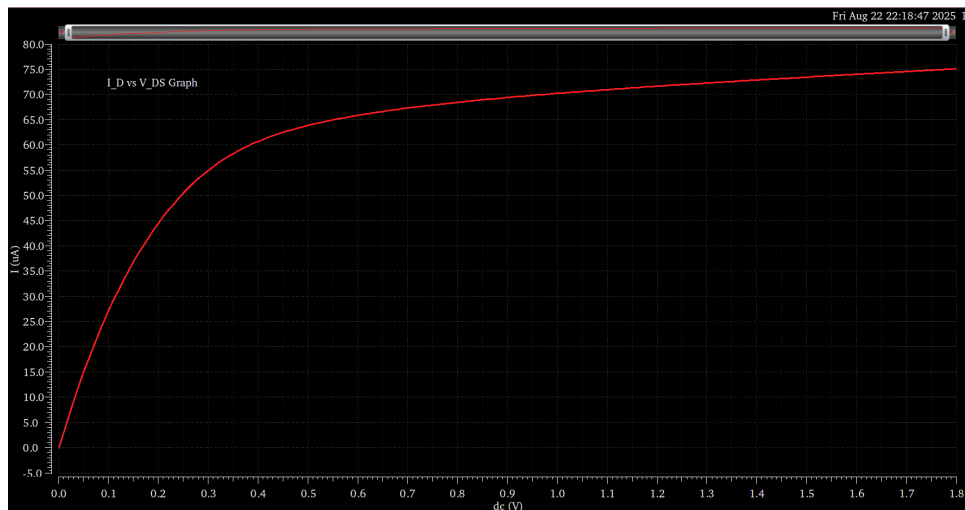


Figure 3: I_D vs V_{DS} for NMOS

2.3 I_D against V_{DS} & V_{GS} for NMOS

As an extension, various values of V_{GS} were used to plot I_D against V_{DS} , resulting in the graph shown in figure 4.

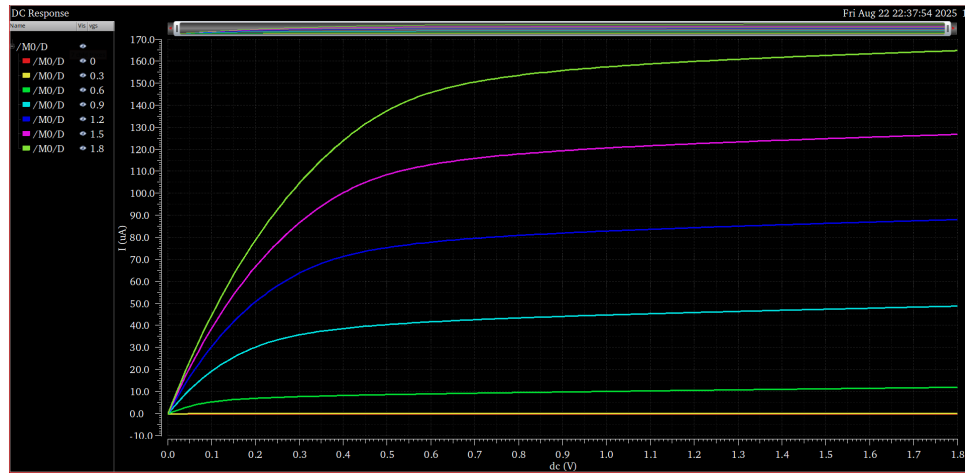


Figure 4: I_D vs V_{DS} for various V_{GS} values

You can see how the value of V_{GS} affects the I_D vs V_{DS} curve, with higher V_{GS} values leading to higher I_D values for the same V_{DS} . Additionally, you can see how the triode region shifts right as the value of V_{GS} increases.

3 PMOS Characteristics

The PMOS circuit was set up similarly to the NMOS circuit, with the PMOS parameters being $l = 180nm$, $w = 220nm$, and $V_{SG} = 1.8V$. The circuit is shown in figure 5.

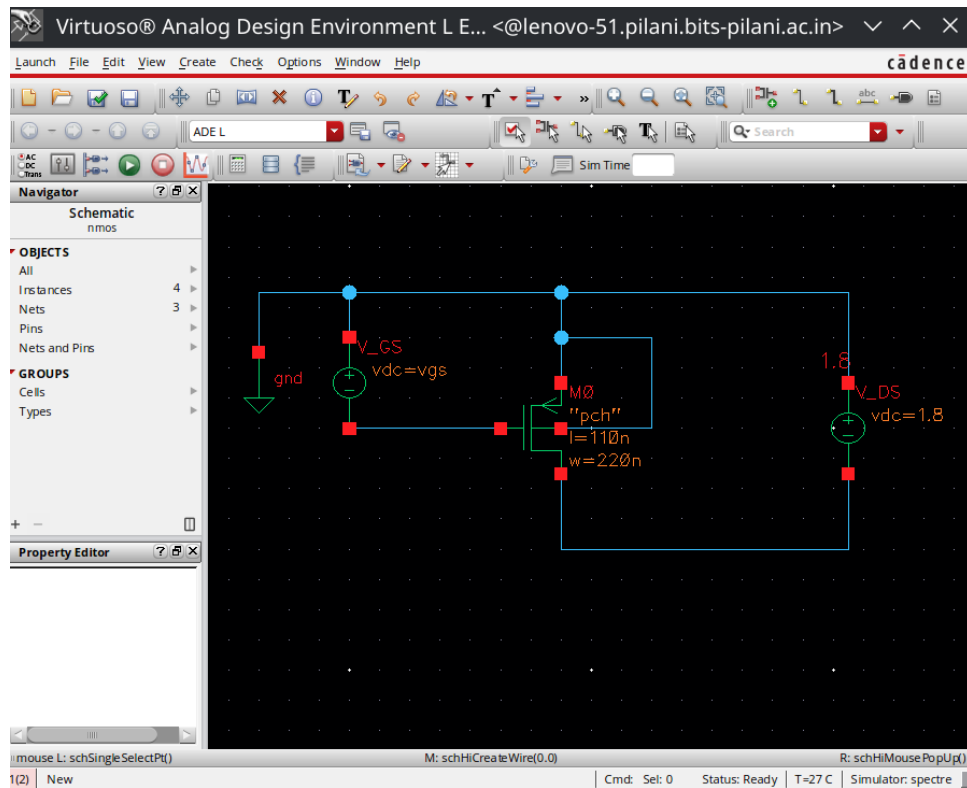


Figure 5: PMOS Circuit

3.1 I_D vs V_{DS} for PMOS

The following graph was obtained for I_D vs V_{DS} for the PMOS circuit from figure 5:

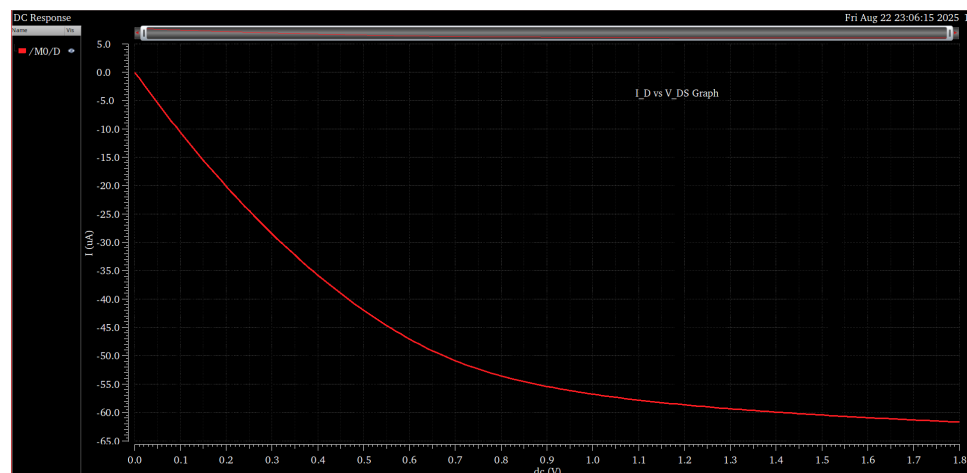


Figure 6: I_D vs V_{DS} for PMOS

3.2 I_D vs V_{GS} for PMOS

The following graph was obtained for I_D vs V_{GS} for the PMOS circuit from figure 5:

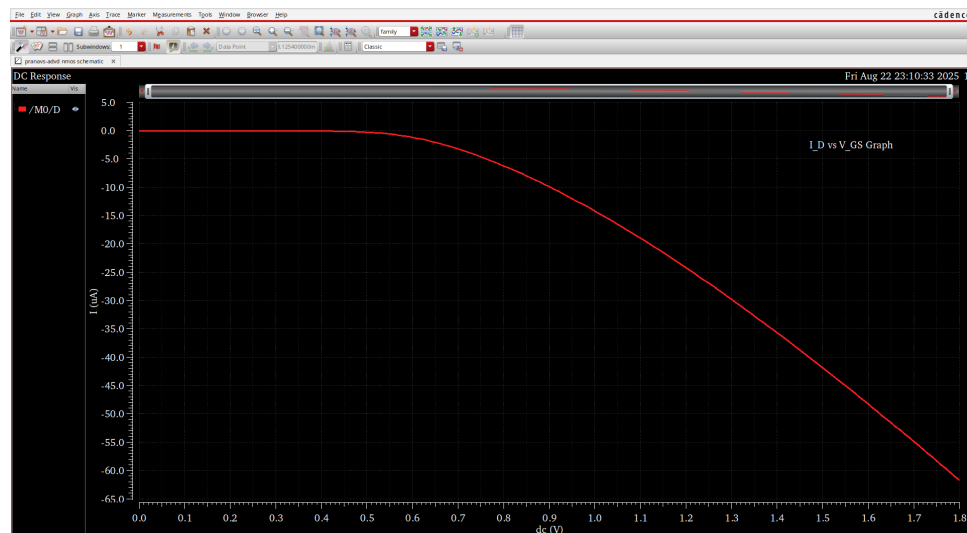


Figure 7: I_D vs V_{GS} for PMOS

You can see the sharp cutoff at a certain voltage of V_{GS} , which indicates the threshold voltage V_{th} .